VENI VIDI VICI

CYBERSECURITY CAPTURE THE FLAG

Challenge Architecture and Technical Briefing

Proposed Structure for PARSEC

6 Cybersecurity Domains | **12 Challenges** | **24–30 Flags**

A practical, scenario-driven cybersecurity experience

Veni Vidi Vici Organizing Team

PARSEC | IIT Dharwad

Technical Briefing | Proposed Architecture

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1 Competition Overview

Veni Vidi Vici (VVV) is the proposed cybersecurity Capture The Flag competition under PARSEC at IIT Dharwad.

The competition is intended to provide participants with a practical security experience that goes beyond isolated puzzles and textbook exercises.

The challenge set will be organized around six major cybersecurity domains, with two primary challenge environments proposed for each domain.

The current target is:

6	**12**	**24–30**
Domains	Challenges	Flags

The exact number of flags will remain flexible until the challenges have been implemented and playtested.

1.1 Competition Philosophy

VVV is designed around the principle:

Think like an attacker, reason like a defender.

Rather than simply asking participants to identify vulnerabilities, the challenge environments should encourage them to:

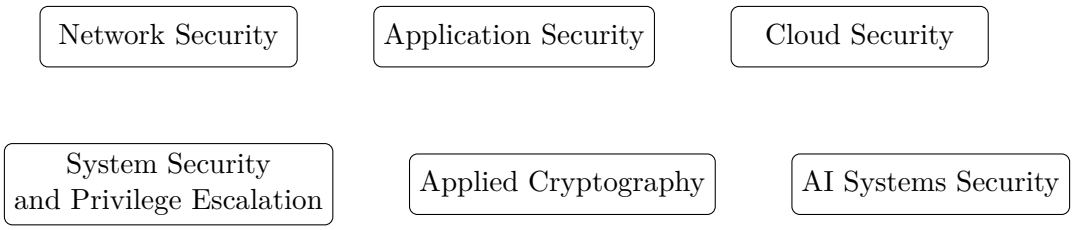
- investigate an unfamiliar system;
- identify its attack surface;
- understand how components interact;
- discover security weaknesses;
- exploit those weaknesses;
- progress through multiple stages where appropriate;
- and understand the security implications of their actions.

2 Challenge Architecture at a Glance

The challenge architecture is organized into three dimensions:

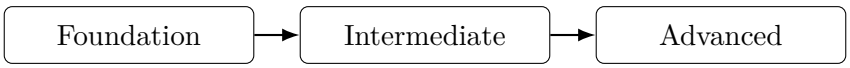
1. **Domain** – what area of cybersecurity the challenge focuses on.
2. **Difficulty** – the level of technical depth required.
3. **Progression** – how participants move from one objective to another within a challenge.

2.1 Six-Domain Architecture



2.2 Challenge Progression

The challenge set is intended to move participants from foundational tasks toward increasingly complex attack reasoning.



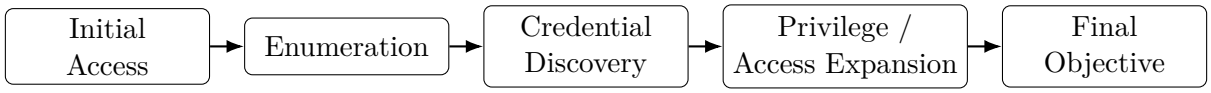
The progression is not intended to imply that every participant must complete challenges sequentially. Participants should be able to select challenges based on their skills and interests.

3 Challenge Structure

The proposed structure consists of twelve primary challenge environments.

Each challenge may contain multiple flags representing meaningful milestones.

For example:



This means that a challenge can provide several meaningful points of progress rather than having a single “all-or-nothing” flag.

4 Proposed Challenge Matrix

The following matrix represents the current architectural baseline.

ID	Domain	Difficulty	Flags	Primary Concepts
N1	**Network Security**	Foundation	2	Reconnaissance, service enumeration
N2	**Network Security**	Intermediate	3	Traffic analysis, internal discovery, pivoting
A1	**Application Security**	Foundation	2	Web and API security fundamentals
A2	**Application Security**	Intermediate / Advanced	4	Multi-stage application exploitation
C1	**Cloud Security**	Foundation / Intermediate	2	IAM, configuration and access control
C2	**Cloud Security**	Advanced	3	Cloud privilege and attack paths
P1	**System Security**	Intermediate	2	Local privilege escalation
P2	**System Security**	Advanced	4	Compromised host and multi-stage escalation
CR1	**Applied Cryptography**	Foundation / Intermediate	1–2	Cryptographic misuse
CR2	**Applied Cryptography**	Advanced	2	Protocol or mathematical weakness
AI1	**AI Systems Security**	Intermediate	2	RAG and AI application security
AI2	**AI Systems Security**	Advanced	3–4	AI agents and tool security

4.1 Target Distribution

The intended distribution is approximately:

Difficulty	Challenges	Purpose
Foundation	4	Accessibility and onboarding
Intermediate	5	Core technical depth
Advanced	3	High-complexity attack chains

This distribution is provisional and may be adjusted after playtesting.

5 Domain I – Network Security

Network Security focuses on networks, exposed services, protocols, traffic, and internal infrastructure.

5.1 Primary Concepts

- Reconnaissance and enumeration
- Port and service discovery
- Service fingerprinting
- Network mapping
- Packet and PCAP analysis
- Protocol analysis
- Insecure protocols

- Authentication weaknesses
- Network segmentation
- Internal service discovery
- Pivoting and lateral movement

5.2 N1 – Network Reconnaissance

The introductory challenge is intended to establish the fundamental workflow of identifying and understanding an exposed attack surface.

Participants may be required to identify services, understand their behavior, and locate an initial security weakness.

5.3 N2 – Internal Network

The intermediate environment introduces the concept that obtaining access to one system does not necessarily represent the end of an attack.

A conceptual progression is:

$$\text{External Service} \rightarrow \text{Initial Access} \rightarrow \text{Internal Enumeration} \rightarrow \text{Pivot} \rightarrow \text{Internal Objective}$$

6 Domain II – Application Security

Application Security focuses on web applications, APIs, authentication, authorization, and application-level trust boundaries.

6.1 Primary Concepts

- Injection vulnerabilities
- Cross-site scripting
- Path traversal
- Server-side request forgery
- Command injection
- Broken access control
- IDOR / BOLA
- Authentication bypass
- Session and JWT security
- File upload vulnerabilities
- Business-logic vulnerabilities

6.2 A1 – Web and API Security

The foundation application challenge introduces participants to application reconnaissance and common security boundaries.

6.3 A2 – Multi-Stage Application Attack

The advanced application environment may combine multiple weaknesses into a single scenario:

Application Exposure → Initial Access → Credential Discovery → Internal Resource → Final Objective

7 Domain III – Cloud Security

Cloud Security introduces identity, access control, resource configuration, and service relationships.

7.1 Primary Concepts

- Identity and Access Management
- Least privilege
- Excessive permissions
- Cloud storage security
- Publicly exposed resources
- Credential and secret leakage
- Service-to-service trust
- Metadata access
- Cloud privilege escalation
- Cloud attack paths

7.2 C1 – Cloud Configuration

The introductory environment focuses on identifying insecure configuration and excessive permissions.

7.3 C2 – Cloud Attack Path

The advanced challenge may combine an application or service compromise with cloud identity and access-control weaknesses.

Cloud environments will be isolated or simulated and will not expose production infrastructure.

8 Domain IV – System Security and Privilege Escalation

This domain focuses on operating-system security and privilege boundaries.

8.1 Linux Concepts

- File permissions
- SUID and SGID
- Linux capabilities
- Sudo configuration

- Cron jobs
- Services
- Environment variables
- Writable files and directories

8.2 Windows Concepts

- Services
- Scheduled tasks
- Permissions
- Credential discovery
- Privilege boundaries

8.3 P1 – Local Privilege Escalation

Participants begin with restricted access and must enumerate the environment before identifying a privilege-boundary weakness.

8.4 P2 – Compromised Host

The advanced challenge may combine:

Initial Foothold → Enumeration → Credential Discovery → Lateral Movement → Privilege Escalation → Object

9 Domain V – Applied Cryptography

Applied Cryptography focuses on practical weaknesses in cryptographic systems rather than simple ciphertext decoding.

9.1 Primary Concepts

- Cryptanalysis
- Symmetric cryptographic misuse
- IV and nonce misuse
- Weak randomness
- RSA weaknesses
- Key management
- Digital signature failures
- Cryptographic protocols
- Implementation weaknesses
- Mathematical reasoning

9.2 CR1 – Cryptographic Misuse

A practical problem involving an incorrectly implemented or incorrectly used cryptographic construction.

9.3 CR2 – Advanced Cryptographic Security

A more demanding challenge involving protocol analysis, mathematical reasoning, or implementation-level weaknesses.

10 Domain VI – AI Systems Security

AI Systems Security is included as a dedicated domain covering security issues in modern AI-enabled applications and systems.

10.1 Primary Concepts

- Prompt and instruction security
- Direct and indirect prompt injection
- Context manipulation
- Retrieval-Augmented Generation security
- Retrieval poisoning
- Vector-store access control
- AI agent security
- Tool and API abuse
- Excessive agent permissions
- AI application security
- Model security
- AI supply-chain security

10.2 AI1 – RAG and AI Application Security

The intermediate environment focuses on the trust boundaries between an AI model, its context, retrieval system, and protected data.

10.3 AI2 – AI Agent Security

The advanced environment may explore the consequences of connecting an AI agent to tools, APIs, files, or backend services.

A conceptual architecture is:

$$\text{LLM} \rightarrow \text{Agent} \rightarrow \text{Tools} \rightarrow \text{Backend Services} \rightarrow \text{Protected Resource}$$

AI challenges should use controlled and sandboxed environments so that model behavior does not make challenge outcomes unpredictable.

11 Real-World Scenario Design

A central objective of VVV is to move away from purely puzzle-oriented challenges toward scenario-driven security problems.

Instead of presenting a participant with only:

"Find the flag."

the challenge should provide a context in which the participant must understand a system and determine how it can be compromised.

For example, a challenge may present:

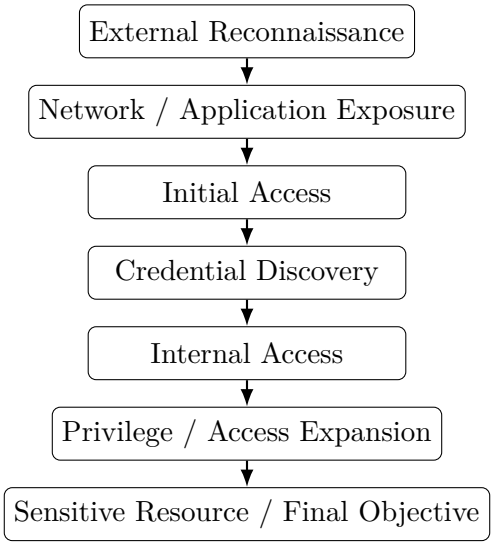
A company has deployed an internal API. An authorization boundary has been incorrectly implemented. Investigate the application, identify the security weakness, and obtain access to the protected resource.

The exact scenario will vary by challenge, but the general principle is that participants should be solving a security problem rather than simply locating an arbitrary string.

12 Cross-Domain Attack Chains

Selected advanced challenges may intentionally combine multiple domains.

A representative attack chain may look like:

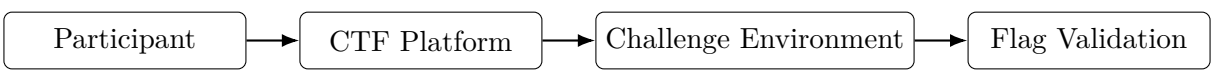


The purpose is to expose participants to the idea that real-world security incidents often involve multiple interacting weaknesses.

13 Technical Infrastructure

The implementation architecture will prioritize isolation, repeatability, and maintainability.

The conceptual participant flow is:



13.1 Infrastructure Principles

The challenge infrastructure should support:

- isolated environments,
- reproducible deployment,
- automated reset,
- deterministic flag validation,
- reusable base environments,
- controlled resource consumption,
- monitoring and troubleshooting.

13.2 Isolation and Safety

All intentionally vulnerable systems should be isolated from production infrastructure.

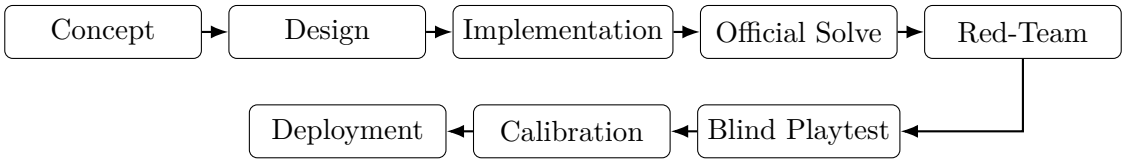
Participants should never receive access to:

- real organizational credentials,
- production cloud accounts,
- production databases,
- unrestricted external systems.

The competition infrastructure should be designed so that participants can safely experiment within the intended challenge boundaries.

14 Challenge Development and Testing

Every challenge should pass through a defined development cycle.



14.1 Testing Requirements

Before deployment, each challenge should undergo:

1. Intended-solution validation.
2. Deployment and reset testing.
3. Unintended-solution testing.
4. Resource and stability testing.
5. Blind playtesting.
6. Difficulty calibration.

15 Industry and Sponsor Integration

Industry participation can significantly improve the realism and relevance of VVV challenges.

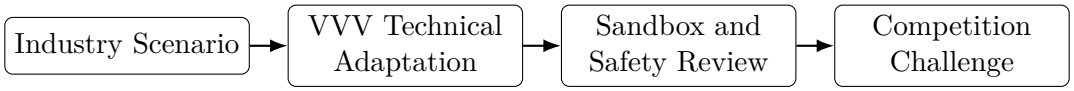
Sponsors may contribute:

- industry-inspired security scenarios,
- application security problems,
- cloud security scenarios,
- network security scenarios,
- incident-response scenarios,
- cryptographic problems,
- AI security scenarios,
- multi-stage attack scenarios.

15.1 Sponsor Problem Statement Model

A sponsor does not necessarily need to provide a complete CTF challenge.

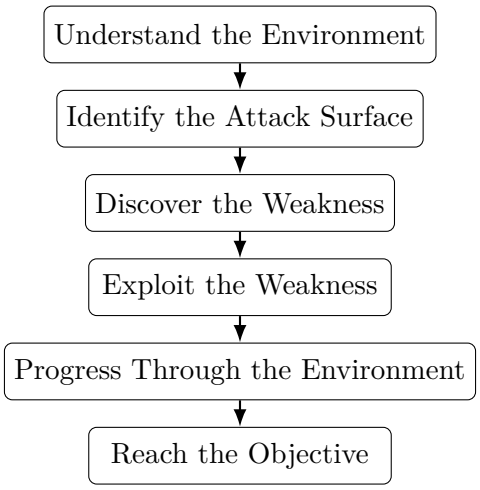
A problem statement, security scenario, vulnerability class, or industry-inspired case can be adapted by the VVV organizing team.



This model allows industry partners to contribute technical expertise without requiring them to build and operate the entire competition infrastructure.

16 Expected Participant Experience

The intended experience can be summarized as:



The desired outcome is for participants to leave with experience in security investigation and reasoning rather than only a collection of individual exploitation techniques.

17 Scope and Design Boundaries

The current scope is intentionally limited to twelve primary challenge environments.

The competition will prioritize:

- technical depth,
- realistic scenarios,
- reliable infrastructure,
- progressive difficulty,
- meaningful flags,
- cross-domain reasoning,
- safe sandboxed execution.

The architecture does not require every challenge to use complex infrastructure.

Instead, infrastructure complexity will be introduced only where it meaningfully contributes to the intended security experience.

18 Architecture Summary

Component	**Proposed Structure**
Cybersecurity Domains	6
Primary Challenges	12
Target Flags	24–30
Difficulty Levels	Foundation, Intermediate, Advanced
Challenge Style	Scenario-driven and hands-on
Advanced Design	Multi-stage and cross-domain attack chains
Cloud Environment	Sandboxed / simulated
AI Environment	Controlled and sandboxed
Infrastructure	Isolated and reproducible
Testing	Internal, red-team, and blind playtesting
Industry Integration	Sponsor-provided scenarios and problem statements

Conclusion

Veni Vidi Vici is envisioned as a practical cybersecurity competition that combines six areas of modern security into a coherent participant experience.

The proposed structure of twelve challenges and approximately 24–30 flags is intended to balance technical depth with the practical constraints of challenge development and infrastructure management.

The central design principle is simple:

Do not just find the vulnerability. Understand the system, exploit the weakness, and reason about the attack.

The architecture is intentionally flexible so that relevant industry-generated problem statements can be incorporated during challenge development.

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